

Problem set 10 - Due: Wednesday, April 29

1. Find the inverse Laplace transform of

$$F(s) = \frac{1}{s^2 - 9}.$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 - 9} \right\} = \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{3}{s^2 - 3^2} \right\} = \frac{1}{3} \sinh(3t).$$

2. Find the inverse Laplace transform of

$$F(s) = \frac{2}{s^2(s^2 + 36)}.$$

Using partial fractions,

$$\frac{2}{s^2(s^2 + 36)} = \frac{A}{s^2} + \frac{Bs + C}{s^2 + 36}.$$

Thus,

$$2 = A(s^2 + 36) + (Bs + C)s^2 = As^2 + 36A + Bs^3 + Cs^2 = (B)s^3 + (A + C)s^2 + 36A.$$

Hence,

$$\begin{aligned} B &= 0, \\ A + C &= 0, \\ 36A &= 2. \end{aligned}$$

So,

$$A = \frac{1}{18}, \quad C = -\frac{1}{18}, \quad B = 0.$$

Therefore,

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{2}{s^2(s^2 + 36)} \right\} &= \mathcal{L}^{-1} \left\{ \frac{1}{18s^2} \right\} + \mathcal{L}^{-1} \left\{ -\frac{1}{18(s^2 + 36)} \right\} \\ &= \frac{1}{18} \mathcal{L}^{-1} \left\{ \frac{1}{s^2} \right\} - \frac{1}{18 \cdot 6} \mathcal{L}^{-1} \left\{ \frac{6}{s^2 + 6^2} \right\} \\ &= \frac{1}{18} t - \frac{1}{18 \cdot 6} \sin(6t) = \frac{1}{18} t - \frac{1}{108} \sin(6t). \end{aligned}$$

3. Solve the following initial value problem using Laplace transform

$$x'' - x' - 6x = 0, \quad x(0) = 2, \quad x'(0) = -1.$$

We have

$$\begin{aligned} \mathcal{L}\{x'' - x' - 6x\} &= \mathcal{L}\{0\} = 0 \\ \implies s^2 X(s) - sx(0) - x'(0) - (sX(s) - x(0)) - 6X(s) &= 0 \\ \implies s^2 X(s) - 2s + 1 - sX(s) + 2 - 6X(s) &= 0 \\ \implies (s^2 - s - 6)X(s) - 2s + 3 &= 0 \\ \implies (s+2)(s-3)X(s) - 2s + 3 &= 0 \\ \implies X(s) &= \frac{2s}{(s+2)(s-3)} - \frac{3}{(s+2)(s-3)}. \end{aligned}$$

Then, by the inverse Laplace,

$$x(t) = \mathcal{L}^{-1}\left\{\frac{2s}{(s+2)(s-3)}\right\} - \mathcal{L}^{-1}\left\{\frac{3}{(s+2)(s-3)}\right\}.$$

Using partial fractions,

$$\frac{2s}{(s+2)(s-3)} = \frac{A}{s+2} + \frac{B}{s-3}.$$

So,

$$2s = A(s-3) + B(s+2).$$

Let $s = -2$, then

$$2(-2) = -5A = -4 \implies A = \frac{4}{5}.$$

Let $s = 3$, then

$$2(3) = 5B \implies B = \frac{6}{5}.$$

So,

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{2s}{(s+2)(s-3)}\right\} &= \mathcal{L}^{-1}\left\{\frac{4}{5(s+2)}\right\} + \mathcal{L}^{-1}\left\{\frac{6}{5(s-3)}\right\} \\ &= \frac{4}{5}\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} + \frac{6}{5}\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} \\ &= \frac{4}{5}e^{-2t} + \frac{6}{5}e^{3t}. \end{aligned}$$

Next,

$$\frac{3}{(s+2)(s-3)} = \frac{A}{s+2} + \frac{B}{s-3} \implies 3 = A(s-3) + B(s+2).$$

Then,

$$3 = -5A \implies A = -\frac{3}{5},$$

and

$$3 = 5B \implies B = \frac{3}{5}.$$

Hence,

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{3}{(s+2)(s-3)}\right\} &= -\frac{3}{5}\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} + \frac{3}{5}\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} \\ &= -\frac{3}{5}e^{-2t} + \frac{3}{5}e^{-3t}.\end{aligned}$$

The full solution is then

$$\begin{aligned}x(t) &= \frac{4}{5}e^{-2t} + \frac{6}{5}e^{3t} - \left(-\frac{3}{5}e^{-2t} + \frac{3}{5}e^{-3t}\right) \\ &= \frac{4}{5}e^{-2t} + \frac{6}{5}e^{3t} + \frac{3}{5}e^{-2t} - \frac{3}{5}e^{-3t} \\ &= \left(\frac{4}{5} + \frac{3}{5}\right)e^{-2t} + \left(\frac{6}{5} - \frac{3}{5}\right)e^{-3t} \\ &= \frac{7}{5}e^{-2t} + \frac{3}{5}e^{-3t}.\end{aligned}$$

4. Solve the following initial value problem using Laplace transform

$$x'' + 4x = \sin(3t), \quad x(0) = x'(0) = 0.$$

We have

$$\begin{aligned}\mathcal{L}\{x'' + 4x\} &= \mathcal{L}\{\sin(3t)\} \\ \implies s^2X(s) - sx(0) - x'(0) + 4X(s) &= \frac{3}{s^2 + 3^2} \\ \implies (s^2 + 4)X(s) &= \frac{3}{s^2 + 9} \\ \implies X(s) &= \frac{3}{(s^2 + 9)(s^2 + 4)}.\end{aligned}$$

By partial fractions

$$\begin{aligned}X(s) &= \frac{3}{(s^2 + 9)(s^2 + 4)} \\ &= \frac{A}{s^2 + 9} + \frac{B}{s^2 + 4}.\end{aligned}$$

Thus,

$$\begin{aligned}3 &= A(s^2 + 4) + B(s^2 + 9) \\ &= As^2 + 4A + Bs^2 + 9B \\ &= (A + B)s^2 + (4A + 9B).\end{aligned}$$

Hence,

$$\begin{aligned}A + B &= 0, \\ 4A + 9B &= 3.\end{aligned}$$

From $A + B = 0$, we have $A = -B$, so

$$\begin{aligned}4(-B) + 9B &= 3 \\5B &= 3 \\B &= \frac{3}{5}.\end{aligned}$$

Thus,

$$A = -\frac{3}{5}.$$

So

$$X(s) = -\frac{3}{5} \frac{1}{s^2 + 9} + \frac{3}{5} \frac{1}{s^2 + 4}.$$

By inverse Laplace,

$$\begin{aligned}x(t) &= -\frac{3}{5} \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 3^2} \right\} + \frac{3}{5} \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 2^2} \right\} \\&= -\frac{3}{15} \mathcal{L}^{-1} \left\{ \frac{3}{s^2 + 3^2} \right\} + \frac{3}{10} \mathcal{L}^{-1} \left\{ \frac{2}{s^2 + 2^2} \right\} \\&= -\frac{1}{5} \sin(3t) + \frac{3}{10} \sin(2t).\end{aligned}$$